CHAPTER

8

UX Research Data Modeling I: User, Flow, Artifact, Physical, and Information Models

Highlights

- UX research data models explained
- User work role model
- User persona model
- Flow (of work) model
- Artifact model
- Physical work environment model
- Information architecture model

8.1 INTRODUCTION

This chapter covers the user, workflow, artifact, physical, and information UX research data models.

8.1.1 You Are Here

We begin each process chapter with a "You are here" picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 8.1). Within the understand-needs UX design lifecycle activity, this chapter is the first of two about the data modeling sub-activity in which you represent in simple models the UX research data you gathered in Chapter 6 and analyzed in Chapter 7.

8.1.2 What Are UX Research Data Models, and How Are They Used?

Modeling is a synthesis technique used to make sense out of your UX research data by creating structured artifacts or representations of raw UX research data to inform UX design. Each model provides a different perspective into the

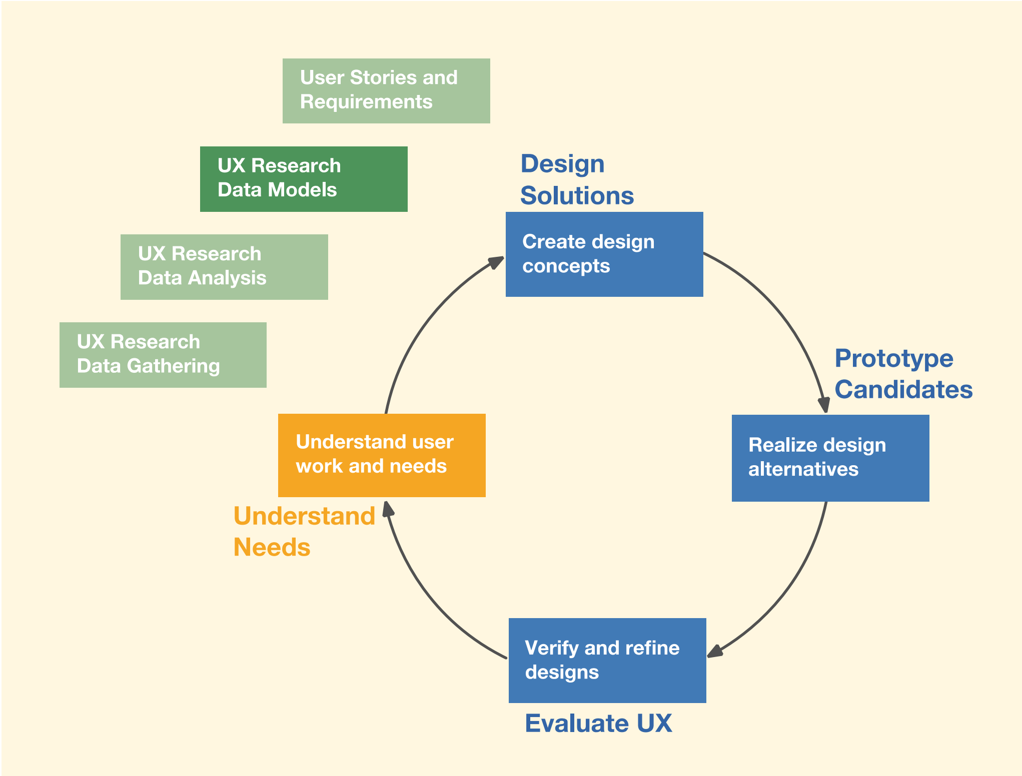

Fig. 8.1

You are here, where we construct UX research data models within the understand-needs lifecycle activity in the context of the overall Wheel UX lifecycle.

overall picture of work[1] practice to synthesize different views of what constitutes the proverbial elephant. Models use abstraction to boil things down to the essence and turn UX research data into actionable items for design. Most of all, the models offer immersive mental frameworks of design parameters.

As a UX design synthesis technique, data models:

- Help clean up the mess that comes with the chaos of raw UX research data (Kolko, 2010)
- Add clarity by setting raw UX research data within specific dimensions describing user work roles, information flow, artifacts, physical work environment, task structures, task sequences, and more

[1]The term *work* broadly signifies anything people/users do while using products or systems (see Section 6.2).

- Help designers focus on the important aspects of design needs by abstracting out the less relevant material
- Aid designers in discovering relationships among the data
- Bring together stories about how an organization functions

8.1.3 Building Data Models Through Synthesis

As you distill raw UX research data into work activity notes (see Section 7.2), or soon thereafter, you will build up the appropriate data models of this chapter and the next through synthesis of the work activity notes (see Section 7.3) and any other inputs available, using descriptions of the models here and in Chapter 9 as a guide.

You will rarely have to construct all these models for every project. As we advise in Section 7.3.8, pick the models that help you best in understanding the work practice you are studying. In Chapter 10, we will derive requirements for the system we are designing from all these models and other inputs from customers and users.

8.1.4 Kinds of Data Models

In this chapter, we present the models in the approximate order of importance and frequency of use:

- The most commonly needed models—in almost every project, you should make user work role models and flow models.
- If there is a broad range of user characteristics for any given user work role, consider making a user persona.
- If there are a large number of different user tasks, organize them with a task structure model.
- If some user tasks are a bit complex, describe them with task sequence models.
- If the work practice is artifact centric, explain it with an artifact model.
- If the work practice is influenced by physical layout, portray that with a physical work environment model.
- If the data and information users need to store, retrieve, display, and manipulate are complex, represent it with an information architecture model.
- The social model (Section 9.4) is needed when social and cultural interactions among the people involved in the work practice are complex and/or problematic.

8.1.5 Modeling Can Pervade the UX Research Process

As you become an efficient UX researcher, you will see modeling opportunities at all stages of UX research. Modeling can begin even before you enter the field for UX research. Starting from the project commission (see Section 5.3), basic model-related information is among the earliest things you learn. The project proposal or business brief that kicks off the project and the early client meetings to define the project provide basic inputs to at least the following:

- User work role model
- Flow model
- Task models

In data gathering, you add to the models and start new ones as you encounter model-related UX research data. As you perform UX research analysis and synthesis especially, modeling becomes a primary focus. Then you will complete and validate the models as we describe here in Chapters 8 and 9, confirming points and answering questions about the data models.

8.2 USER WORK ROLE MODEL

The work role model is one of the most important models, and every project needs one. This model, at its base, is a simple representation of user work roles, sub-roles, and associated user class characteristics. It is essential to identify the key user work (or play) roles as early in UX research as possible.

8.2.1 What Is a User Work Role?

A user work role is not a person, but a work assignment defined by the duties, functions, and work activities of a person with a certain job title or responsibility, such as customer or database administrator. Job titles themselves, however, do not necessarily make good names for user work roles; instead, use names that distinguish them by their respective types of work.

For example, for the Middleburg University Ticket Transaction Service (MUTTS), the existing system, the ticket seller role is filled by someone who helps customers buy tickets. A person in that role does entirely different tasks than, say, the event manager who, behind the scenes, enters entertainment event information into the system so that tickets can be printed, offered, and purchased.

Because roles are defined by the work, different people can fulfill the same work role. For example, multiple people at a bank can assume the single work role of a bank teller.

Among the other MUTTS roles we discovered early in UX research are:

- Event manager, who negotiates with event promoters about event information and tickets to be sold by the MUTTS ticket office
- Advertising manager, who negotiates with outside advertisers to arrange for advertising to be featured via MUTTS (e.g., ads printed on the back of tickets, posted on bulletin boards, on the website)
- Financial administrator, who is responsible for accounting and credit card issues
- Maintenance technician, who maintains the MUTTS ticket office computers, website, ticket printers, and network connections
- Database administrator, who is responsible for the reliability and data integrity of the event database
- Administrative supervisor, who oversees the entire MU services department
- Office manager, who is in charge of the daily MUTTS operation
- Assistant office manager, who assists the office manager

8.2.2 Work Roles Transcend Job Duties

As we saw in Section 6.2, in UX we use the terms work, work activity, work practice, and work domain to signify an even broader meaning than a role defined by job duties. Work, in this context, embraces all things users do, including activities beyond work and play, and a broad range of human activities to which UX design applies. Work is anything people/users do while using products, systems, devices, or other technology.

8.2.3 Sub-Roles

For some work roles, it can be useful to distinguish sub-roles defined by the different subsets of tasks that people in the work role are trained and authorized to do. Examples of sub-roles for a bank teller role include handling international currency exchange and wire transfers and performing daily checking transactions.

8.2.4 Mediated Work Roles

A work role can:

- Involve direct system usage or not
- Be internal or external to the organization, as long as the role entails participation in the work practice of the organization

Users serving work roles that do not use the system directly, but who are helped by people in roles that do use the system, are called mediated users because their interaction with the system is mediated by direct users. Cooper

(2004) calls them "served users," and they play a major part in the workflow and usage context. As an example, a MUTTS ticket buyer, not a direct user of the system, is helped by a ticket seller, who uses the system directly.

Mediated roles are often customers or clients of the enterprise on whose behalf direct users, such as clerks and agents, conduct transactions with the computer system. They may include point-of-sale customers at a retail outlet or customers needing services at a bank or an insurance agency. The working relationship between the mediated users and the agent is critical to the resulting user experience.

Example: Work Roles in MUTTS

The two obvious work roles in MUTTS (the old system) already mentioned are:

- Ticket buyer, who interacts with the ticket seller to learn about event information and buy event tickets, is a mediated work role.
- Ticket seller, who serves ticket buyers and uses the system to find and buy tickets on behalf of ticket buyers, is a direct work role.

Exercise 8.1 Identifying User Work Roles for Your Product or System

Goal: Practice identifying work roles from your work activity notes.
Activities: By now, you should be fairly certain about the work roles for your system. Using your user-related work activity notes, identify and list the major work roles for your product or system. For each role, add explanatory notes describing the role and a description of the kinds of tasks that people in that role would be expected to perform.
Deliverables: A written list of work roles you identified for your system, each with an explanation of the role and a brief high-level description of the tasks to be performed by this user work role.
Schedule: About 30 minutes should be enough to get the most out of this assignment.

8.2.5 User Class Definitions

A user class for a work role or sub-role is defined by a description of relevant characteristics of the potential user community intended to perform that role. Every work role and sub-role will have at least one user class.

User class definitions can be defined in terms of characteristics, such as demographics, skills, knowledge, and special needs. Some specialized user classes, such as soccer mom, millennial, metrosexual, or elderly citizen, may be dictated by marketing (Frank, 2006).

As an example of a user class, the set of people in the town-resident subrole of the ticket-buyer role in the new Ticket Kiosk System may be expected to include inexperienced (first-time) users from the general public. Another user class for this work role could be senior citizens with limited motor skills and some visual impairment.

8.2.5.1 Knowledge and skill characteristics

User class characteristics related to knowledge and skills include:

- Background, experience, training, education, and/or skills expected in a user to perform a given work role. For example, a given class of users must be trained in retail marketing and must have five years of sales experience
- Knowledge of computers, both in general and with respect to specific systems
- Knowledge of the work domain; that is, knowledge of and experience with the operations, procedures, and semantics of the various aspects of the application area the system being designed is trying to address

For example, a medical doctor may be an expert in domain knowledge related to a magnetic resonance imaging (MRI) system but may have only novice to intermittent knowledge in the area of related computer applications. In contrast, an administrator in the hospital may have little overall domain knowledge about MRI systems but may have more complete knowledge regarding the day-to-day use of computer applications.

Some knowledge- and skills-based characteristics of user class definitions can be mandated by organizational policies or even legal requirements, especially for work roles that affect public safety.

8.2.5.2 Physiologic characteristics

User class characteristics related to physiologic characteristics and accessibility issues include:

- Impairments, limitations, and other Americans with Disabilities Act-related considerations
- Age. If older adults are expected to take on a given work role, then they may have known characteristics to be accommodated in design (e.g., they can be susceptible to sensory and motor limitations that come naturally with age).

Example: User Class Definitions for MUTTS

Ticket seller. Minimum computer skill requirements may include the ability to point and click. Probably some simple additional domain-specific training is necessary. When we interviewed ticket sellers at MUTTS, we discovered that they once had a manual that outlined job responsibilities, but over time it had become outdated and eventually lost.

Because ticket sellers are often hired as part-time employees, there can be considerable turnover. So, as a practical matter, much of the ticket seller training is picked up on the job or while apprenticing with someone more experienced. Expect some mistakes along the way. This variability of competence in the work role, which is the main interface with the public, is not always the best for customer satisfaction, but there does not seem to be a way around it at MUTTS.

Exercise 8.2 User Class Definitions for Your Product or System

Goal: Practice defining user classes for work roles, similar to the previous example.

Activities: Taking your user-related UX research data notes, create a few user class definitions to go with the work roles you identified in Exercise 8.1. Describe the characteristics of each user class.

Deliverables: A few user class definitions to go with the work roles identified.

Schedule: About 30 to 45 minutes should be enough to get the most out of this assignment.

8.2.6 Post the Work Role Modeling Results

Post a visual representation of the updated and refined work roles, sub-roles, and user classes in a central location within the design studio so everyone can refer to it during design.

8.3 USER PERSONAS

A persona, or user persona, is a narrative description of a hypothetical but specific character as a UX design target for a user in a specific work role (Cooper, 2004). A user persona is a technique for making users real to designers. A persona does not represent an actual user, a typical user, or an average of users. It is a description of a realistic but "hypothetical archetype" (Cooper, 2004) person that must be served by the design. It is an individual who has a name, a life, and a personality, allowing designers to limit design focus to something very specific. Each persona applies to exactly one user class of one work role. Creating a user persona is a good example of UX synthesis—bringing multiple instances of raw data together to create a useful input for designers.

It is not possible to make a single design that is the best for everyone who might fill the corresponding work role. If you try, you will often end up with a lowest common denominator design that does not work well for anyone. A persona presents a specific design target that, hopefully, fits the target user type extremely well and will suffice for other users.

8.3.1 Why Personas?

The value of personas to designers is that, in the face of multiple competing possible kinds of users, it allows them to focus on just one design target. Once the proper optimal user persona is created, questions such as "What about this other user?" are easily answered, at least in design discussions about a persona named Charlie, by "Well, that user isn't Charlie."

8.3.2 UX Research Data for Personas

During UX research synthesis, any work activity note that says anything about the user as a person, including personality and habits and how that person uses the product or system, is a good candidate for consideration to be synthesized in a user persona model.

Example: Sample Work Activity Notes Suitable as Inputs to User Personas

Here are a few example work activity notes that are candidates for inputs to user personas in the Ticket Kiosk System:

- *I usually work long hours in the lab on the other side of campus.*
- *I like classical music concerts, especially from local artists.*
- *I love the sense of community in Middleburg.*
- *Sometimes I need to buy a set of tickets with adjacent seating because I like to sit with friends.*
- *I like to buy MU football game tickets in a group so I can sit with my friends.*
- *If I find a ticket for an event I would like to go to, I will call my friends before buying tickets.*
- *If in Middleburg, I would probably just go to the venue to buy tickets; otherwise, I would buy online.*
- *I would use the kiosk if it were affiliated with MU, and they send an email informing us about it.*
- *In a big city, I will be spending more time commuting, so if there is a kiosk in the Metro train station, then I will be more likely to use it.*

- *I have used a kiosk in grocery stores, retail stores with self-checkout, and to pick up tickets at a movie theater, so I am very familiar with this kind of interaction.*
- *Identity theft and credit card fraud are huge concerns to me.*

The main discussion about constructing personas and using them to guide design is in Chapter 13 on generative design.

8.3.3 A Preview of How to Create Personas

Starting with UX research data, you first create multiple candidate personas for each user work role. Each persona is a description of a specific individual who is given a name, a life, a personality, and a profile as a person, especially with respect to how they use the new product or system. Making personas precise and specific is paramount. Specificity is important because that is what lets you exclude other cases when it comes to design.

The most difficult part of creating the persona is choosing the correct one for the design. Through a process to be described in Section 13.3.3.2, you will choose one of those personas as the primary persona, the single best design target, the persona to which the design will be made specific. The trick is to make the target design to be just right for your chosen persona and to suffice for the others.

Example: Personas for the Ticket Kiosk System

An example of a persona for the student sub-role in the Ticket Kiosk System could be that of Jane, a junior-year biology major who is a second-generation MU attendee and a serious MU sports fan with season tickets to MU football games. Jane is a candidate to be the primary persona because she is representative of most MU students when it comes to MU school spirit. Jane's boyfriend, Jeff, is a music major interested in the arts.

Exercise 8.3 Early Sketch of a User Persona for Your Product or System

Goal: Get some experience at writing a persona.
Activities: Select an important work role within your product or system. It is best if at least one user class for this work role is broad, with the user population coming from a large and diverse group, such as the general public. Using your user-related UX research data, create a persona, give it a name, and provide a photo to go with it. Write the text for the persona description.
Schedule: About one hour should be sufficient to complete this exercise.

8.4 THE FLOW MODEL

The flow model is another of the most important models, and every project needs one.

8.4.1 What Is a Flow Model?

The flow model, at its base, is a simple graphic representation of how information, work products, and artifacts flow among user work roles and parts of the product or system as the result of user actions. For example, what is the workflow associated with searching for, purchasing, and downloading a song or piece of music to a personal device?

A flow model is a bird's-eye picture of the work domain, its components, and interconnections among them. It is a high-level operational view of how users in each work role and other system entities interact and communicate to accomplish work. A flow model is especially about how work gets handed off between roles—the places where things are most likely to fall through the cracks. As Beyer and Holtzblatt (1997, p. 236) put it: "The system's job is to carry context between roles."

8.4.2 Central Importance of the Flow Model

Because the flow model is a unifying representation of how the system fits into the workflow of the enterprise, it is important to understand it and get it established as early as possible. Along with the user work role model, the flow model is the centerpiece of immersion (a form of deep thought and analysis of the problem at hand) in your UX design studio. Even though your early conceptualization of workflow may be incomplete and not entirely accurate, you can refine it as a clear picture of the work domain, system, and users slowly emerges. If necessary, you should go back to your users and ask them to verify the accuracy and completeness of the flow model.

8.4.3 How to Make a Flow Model

Starting very early in the UX lifecycle:

- Draw the evolving flow model diagram as a graph of nodes and arcs.
- Start by drawing people icons, labeled with the work roles, as nodes. Include roles external to the organization.
- Add nodes for other entities, such as a database into which and from which anything related to the work practice can flow.

- Draw directed arcs (arrows) representing flow, communication, and coordination necessary to do the work of the enterprise between nodes.
- Label the arcs with what (e.g., artifact, information) is flowing and by what medium (e.g., email, phone calls, letters, memos, and meetings).
- In UX research synthesis, as you encounter work activity notes that describe how work flows in the organization, extract them as inputs to the flow model or merge them directly into the evolving flow model.

Flow models also include non-UI software functionality when it is part of the flow (e.g., the payroll program must be run before printing and then issuing paychecks). If you make a flow model of how a website is used in work practice, do not use it as a flowchart of pages visited; instead, it should represent how information and commands flow among the sites, pages, backend content, and users to carry out work activities.

Sometimes you have to make your flow model very detailed to get at important specifics of the flow and to unearth important questions regarding who does what in depth. Post a large image of the flow model prominently in your UX design studio as a central part of immersion.

Example: Sketching the Flow Model for MUTTS

The simple early MUTTS flow model (Fig. 8.2) is centered on the ticket-buying activity between ticket seller and ticket buyer roles, based on sketches made in data gathering and related work activity notes encountered in UX research synthesis.

Interaction between the ticket buyer and ticket seller may begin with a question about what events are available. The ticket seller then sends a suitable request for this information to the event database, and the information in a response flows back to the ticket seller, who then informs the ticket buyer. After one or more such interactions to establish which events are desired by the ticket buyer, a ticket purchase request flows from ticket buyer to ticket seller and to the event database. The transaction is consummated with payment, and the request then flows to the ticket printer. Printed physical tickets flow to the ticket seller and are then given to (flow to) the ticket buyer.

Example: Extending the MUTTS Flow Model

As more work activity notes related to the flow of the ticket-buying process emerge in work activity notes during UX research synthesis, we can extend and refine the MUTTS flow model. For instance, we added an online ticket source, Tickets4ever.com, partnering with MUTTS (Fig. 8.3).

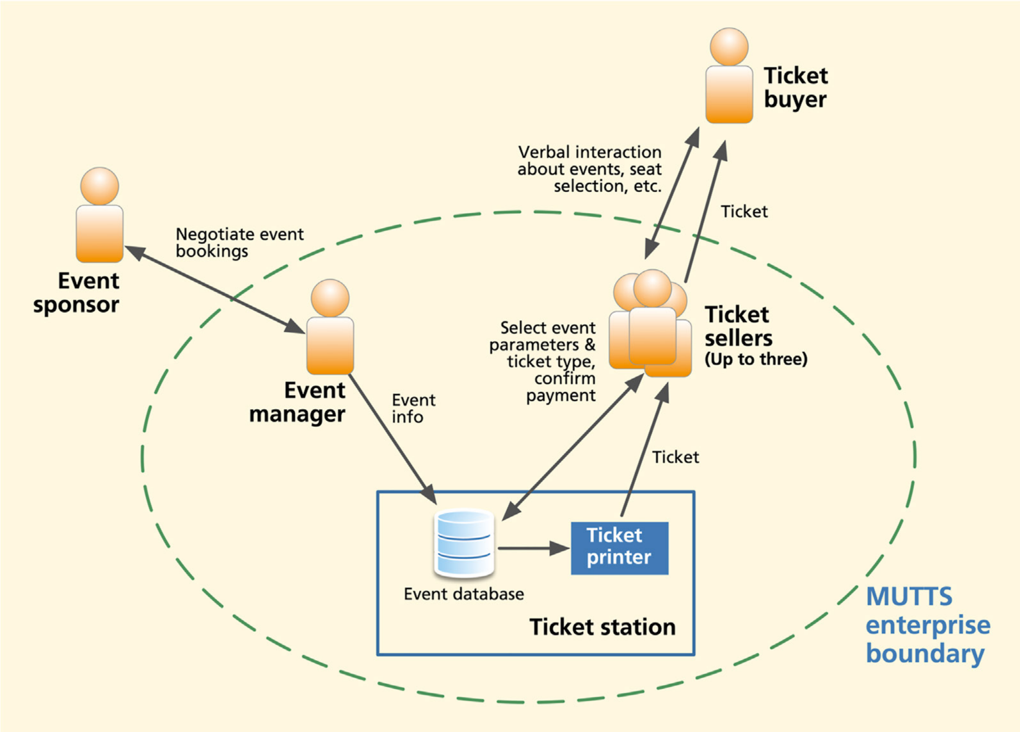

Fig. 8.2

Simple early flow model for the existing MUTTS.

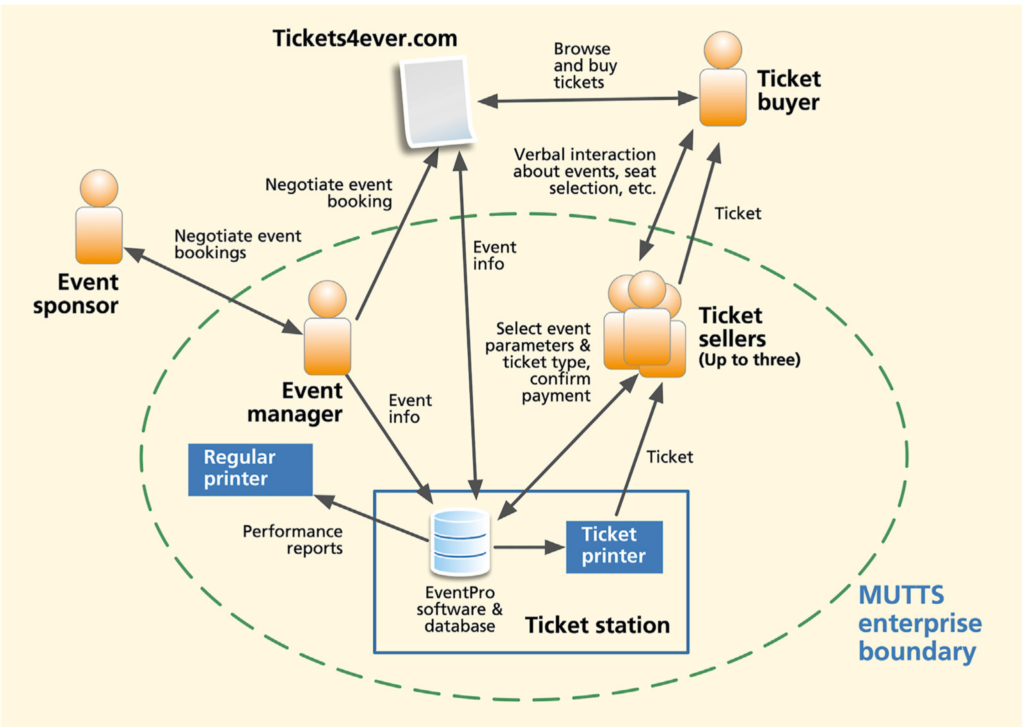

Fig. 8.3

A further step of the flow model sketch of the MUTTS system, showing Tickets4ever.com as a node.

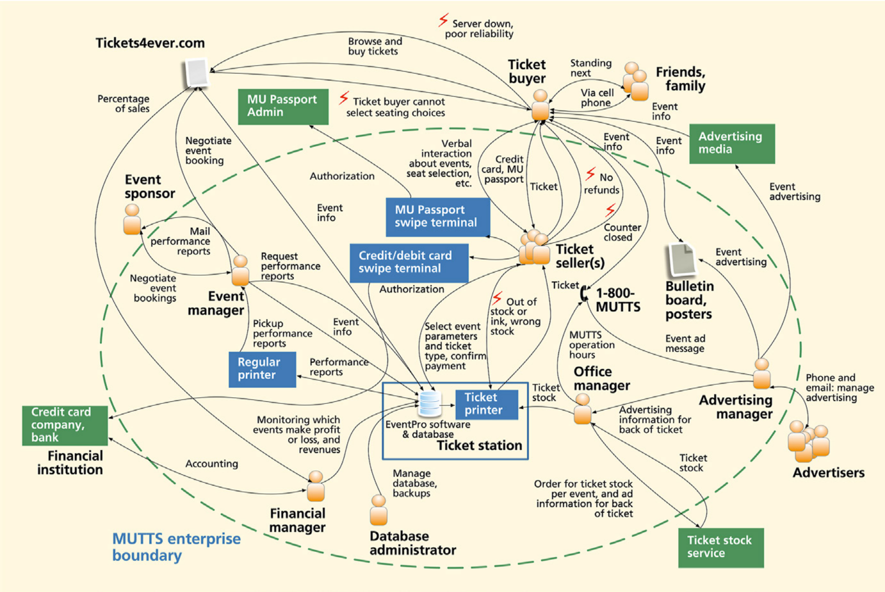

Fig. 8.4

Flow model of our version of MUTTS.

Eventually, the flow model for MUTTS evolved into a rather full diagram (Fig. 8.4). While this is obviously rather complete, it is also too complex to consider as a whole. Design of selected parts will center on the relevant chunks of this graph.

Note the interactions among roles not involved directly in ticket buying or selling, such as friends and/or family of the ticket buyer (in the upper righthand corner of Fig. 8.4) accompanying the buyer in-person or on the cell phone, communicating about choices of events, dates, seats, and prices.

Exercise 8.4 Creating a Flow Model for Your Product or System

Goal: Practice making an initial flow model sketch for the work practice of an organization.

Activities: For your target system, sketch out a flow model diagram in the same style as our flow model sketch for MUTTS, showing work roles, information flow, information repositories, transactions, etc. Draw on your raw UX research data and construct a representation of the flow of data, information, and work artifacts. Begin by representing your work roles as labeled nodes, then add in any other nodes for databases, and so on.

Use labeled arcs to show flow, including flow of information and physical artifacts:

- Show all communication, including direct conversations, email, phones, letters, memos, meetings, etc.
- Show all coordination necessary to do the work of the enterprise. Include both flow internally within the enterprise and flow externally with the rest of the world.

If you do not have enough UX research data from your limited data-gathering exercise, look up other comparable similar business practices to make this work.
Deliverables: A one-page diagram illustrating a high-level flow model for the existing work process of your target product or system.
Schedule: Given a relatively simple domain, we expect you can do this exercise in about one hour.

8.4.4 The Customer Journey Map, a Kind of Flow Model

In Section 6.7.3, we described collecting data about extended usage by shadowing users and documenting the customer journey, the story of an interaction of a user or customer with a product or system and its entire ecology (the surrounding parts of the world, including networks, other users, devices, and information structures, with which a user, product, or system interacts) over the duration of the relationship. This kind of data can be represented by a customer journey map, which is a visual ecologic view of the flow of the customer or user experience over time and through space. A customer journey map tells a story of usage to clients and helps UX designers understand their special work practice and needs.

8.5 ARTIFACT MODEL

An artifact (work artifact) is an object, usually tangible, that plays a role in the workflow of a system or enterprise (e.g., a printed receipt from a restaurant). Not all work practice is artifact centric, but if it is, then work artifacts are among the most important entities that pass from one work role to another within the flow model. What artifacts do users employ, manipulate, and share as part of their work practice? Now is the time to take the artifacts (e.g., paperwork) involved in product or system usage that you collected in data gathering and incorporate them into a simple artifact model.

8.5.1 What Does an Artifact Model Contain?

At this point, your artifact model is probably just a collection of labeled artifacts plus some notes about them. Examples of artifacts include:

- Work practice forms
- Sketches
- Props
- Memos
- Significant email messages
- Correspondence templates
- Product change orders
- Order form
- Receipt
- Paper or electronic forms
- Templates
- Physical or electronic entities that users create, retrieve, use, or reference within a task and/or pass on to another person in the work domain
- Photos (with permission) of the workplace and work being performed
- Other objects that play a role in work performed

Example: Work Artifacts From a Local Restaurant

One of the project teams in our user experience class designed a system to support a more efficient workflow for taking and filling food orders in a local restaurant, but part of a regional chain, using handheld tablet computers. As part of their data gathering, they gathered a set of paperwork artifacts, including manually created order forms, guest checks, and a receipt (Fig. 8.5).

These artifacts are great conversational props as your team interviews those in the different roles that use them. They provide avenues for discussion because almost every restaurant uses these artifacts repeatedly. What are things that work with this kind of artifact for order taking? What are some breakdowns? Is there a way the customer can see what the wait-staff puts down as the order to avoid errors? How does a person's handwriting impact this part of the work activity? What is the interaction like between the wait-staff and the restaurant's guests? What is the interaction like between the wait-staff and the kitchen staff?

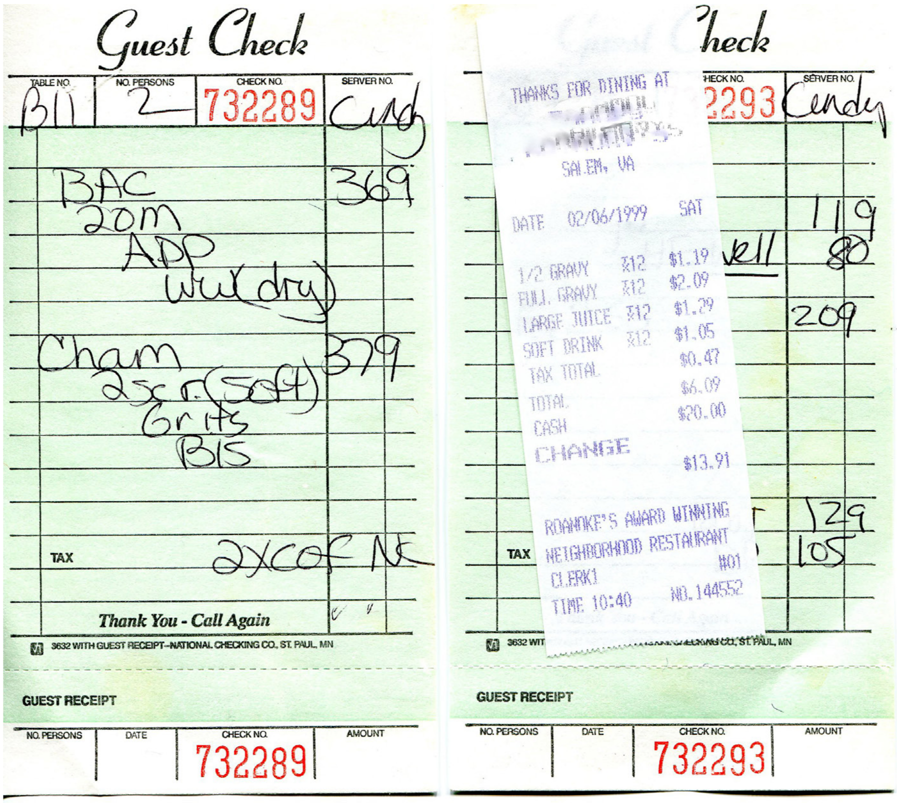

Fig. 8.5

Examples of work artifacts gathered from a local restaurant.

8.5.2 Constructing the Artifact Model

How do you make the artifact model? Well, you now have a collection of artifacts that you gathered in UX research data gathering, including sketches, copies of paperwork, photographs, and real instances of physical work practice artifacts.

Attach samples of each artifact and photos of their use to poster paper (e.g., a blank flip chart page) as exhibits for discussion and analysis. From your data notes, add annotations to these exhibits to explain how each artifact is used in the work practice. Add stick-on notes associating them with tasks, user goals, and task barriers.

These posters are tangible and visual reminders of user work practice that everyone can walk around, think about, annotate with notes and insights, and talk about and, by themselves, are a centerpiece of immersion and the focus of UX synthesis.

But the real importance of work practice artifacts is in how they tell a story within the overall flow. How seamlessly does a given artifact support the flow? Does it cause a breakdown? Is it a transition point between two systems or roles? In this regard, artifacts are an essential component of the flow model.

Example: Combining the Artifacts Into a Restaurant Flow Model

It is easy to think of artifacts associated with the flow model of a restaurant. The first artifact encountered by a person in the customer work role, delivered by the person in the wait-staff work role, is a menu, used by the customer work role to decide on something to order.

Other usual restaurant work artifacts include the order form on which the wait-staff person writes the original order and the guest check (see Fig. 8.5), which can be the same artifact or a printed check if the order is entered into a system. Finally, there might be a regular receipt and, if a credit card is used, a credit card signature form and a credit card receipt. Artifacts in restaurants, as in most enterprises, are the basis for at least part of the flow model. In Fig. 8.6, you

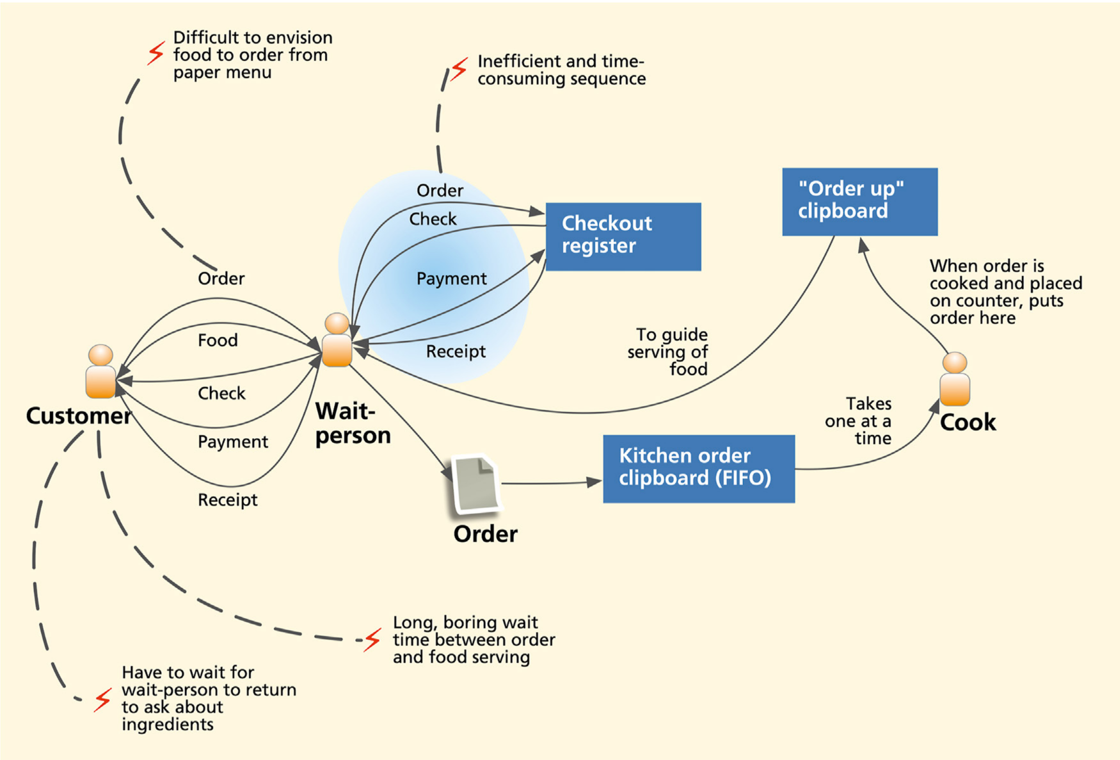

Fig. 8.6

Early sketch of a restaurant flow model with focus on work artifacts derived from the artifact model.

can see how restaurant artifacts help show the workflow from order, to serving, to concluding payment.

The artifacts, especially when arranged as part of a flow model, can help make a connection from UX research data to thinking ahead about design. For example, the waiting and boredom shown in the figure pose a barrier to the customer, represented by a red lightning bolt (ϟ), but that waiting time is also a design opportunity. Providing music, news, entertainment, or even a game to play (e.g., through displays in the tabletop) can alleviate the boredom and gain a competitive advantage in the market.

Example: Artifact Model of Car Keys and Orders in an Auto Repair Shop

As a non-restaurant example of artifacts in workflow, consider the keys to a customer's car as a central artifact in the work practice of an auto repair facility. First, the keys may be put in an envelope with the work order, so the mechanic has them when needed.

After repairs, the keys are hung on a peg board near the customer checkout until the invoice is presented to the customer and the bill paid. To support immersion, you should make yourself a mockup of the peg board with some old keys and add them to the growing collection of artifacts in the model.

8.6 PHYSICAL WORK ENVIRONMENT MODEL

Physical models are pictorial representations of the layout of work locations, personnel, equipment, hardware, the ecology, communications, devices, and physical artifacts that are part of the work practice. The physical arrangement of entities in the flow can affect the outcome of the work. For example, the location of desks on a trading floor in a financial institution could be important because of the many exchanges of information that happen through sign language and other verbal communications among traders.

Merge into the evolving physical model any relevant notes, sketches, diagrams, and photos you made during your data-gathering visits. Other things to include are:

- Floor plans (not necessary to scale)
- Where people and important objects are located and move during interaction
- Locations of:
  - Furniture

- Office equipment (telephones, computers, copy machines, fax machines, printers, scanners)
- Communications connections
- Workstations
- Points of contact with customers and the public

As an example, show in a restaurant where the customer tables are and where the wait-staff go to submit and pick up orders. Make posters of the diagrams and photos, annotated with notes about the physical layout and any problems or task barriers it imposes due to distances, awkward layouts, and physical interference among roles in a shared workspace. Post them in your design studio.

8.6.1 Include Environmental Factors, When Appropriate

When creating physical models, also think of the physical characteristics of a workplace that can affect work activities and add them as annotations. For example, a steel mill floor is about safety concerns, noise, dust, and hot temperatures—conditions where it is difficult to think or work. A system with a terminal on a factory floor means dirty conditions and no place to hold manuals or blueprints. This may result in designs where the use of audio could be a problem, needing more prominent visual design elements for warnings, such as blinking lights.

Example: Physical Model for MUTTS

In Fig. 8.7, we show the physical model for MUTTS. The center of workflow is the ticket counter, containing up to three active ticket seller terminals. On the back wall, relative to ticket sellers, are the credit card and MU Passport card swiping stations. This central ticket-selling area is flanked with the manager's and assistant manager's administrative offices.

Barriers not shown in this figure include one to the ticket buyer lines. At peak times, customers may have to wait in long lines outside the ticket window. The scanner in the manager's office, used to digitize graphic material, such as posters or advertisements for website content, presents barriers to usage because it is very slow and not in a convenient location for all to share.

The ticket printers can also introduce barriers to workflow. Because they are specialized printers for ticket stock, when one goes down or runs out of paper or ink, the employees cannot just substitute another printer. They have to wait until the technician can get it serviced or, worse, until it can be sent out for service.

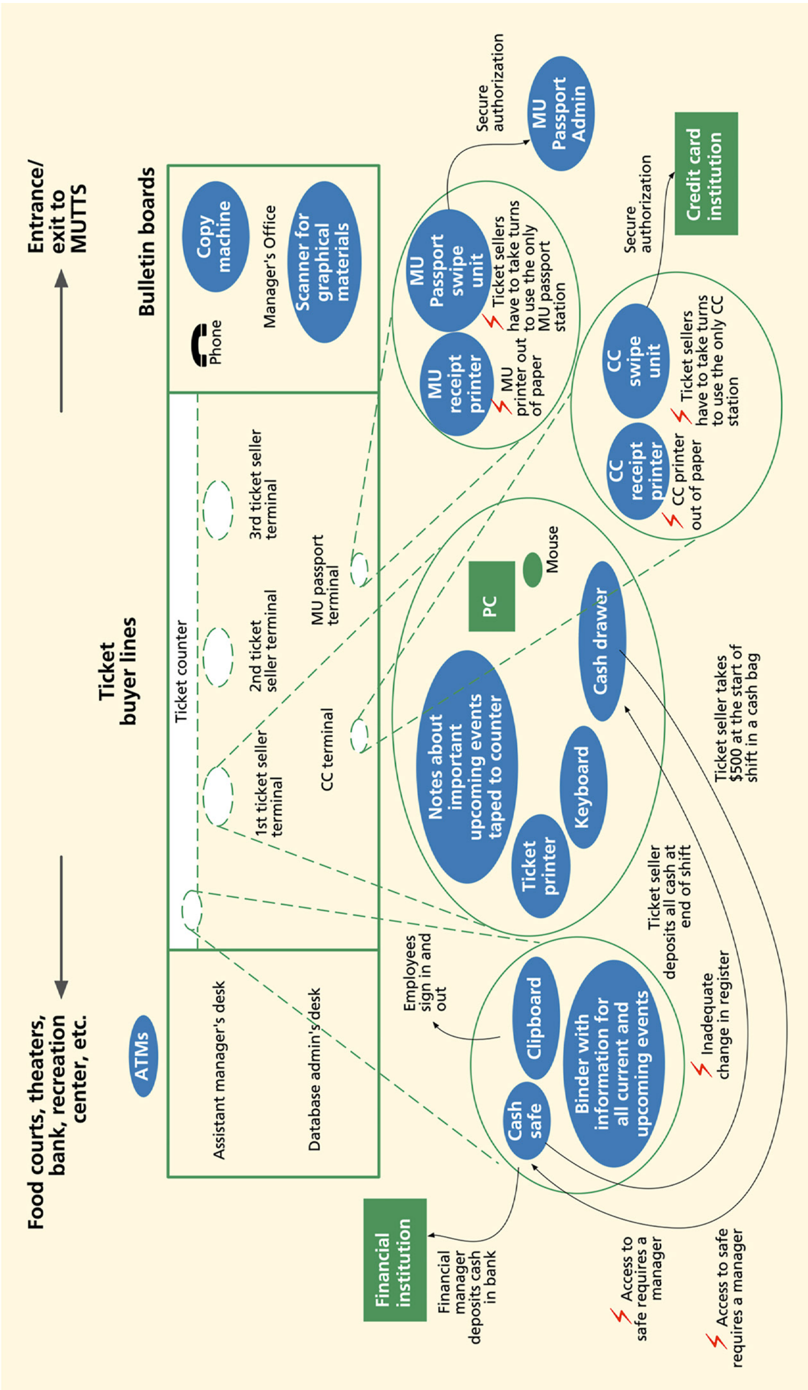

Fig. 8.7
A physical model for MUTTS.

8.7 INFORMATION ARCHITECTURE MODEL

Information architecture is a complete technical description and design of information structures for organizing, storing, retrieving, displaying, manipulating, and sharing information, including designs for labeling, searching, and navigating information. What information do users use and interact with as part of their work, and how is it structured? If this kind of information is complex, represent it with an information architecture model.

Information architecture is usually fairly simple in the context of a domain-simple consumer product but could be quite complex in an enterprise system. In projects where, for example, a database is central to workflow and/or data definitions, and data relationships are crucial to understanding the work, information architecture can be essential to understand early.

Simple information structures can be represented as lists of information objects and their attributes. For example, events are important objects in MUTTS and the Ticket Kiosk System, and they may have attributes, such as event name, event type, and event description. More complex structures, such as patient health records, involve database schemas and entity-relationship diagrams, which tie together usage and design, and are the basis for information displays and data field layouts in wireframes.

Example: Information Architecture Modeling

Consider information about events in the Ticket Kiosk System database. A general event record may have attributes such as the following:

- Event name
- Event type
- Event description
- Range of dates event is occurring
- Ticket costs:
  - Seat types and costs
  - Reserved status
- Venues:
  - Location
  - Capacity
  - Directions to venues
- Video trailers

- Photos
- Reviews

There may also be a many-to-many relationship between events and ticket buyers, including these attributes:

- Date of reservation
- Seat number
- Cost of ticket, as paid

8.8 SEGUE

In this chapter we have started the essential synthesis activity of modeling user roles, work flow, work artifacts and their role in the broader work context, the physical environment of the work practice, and the nature of information as it permeates the fabric of work. We will conclude modeling of other aspects of work such as tasks in the next chapter.